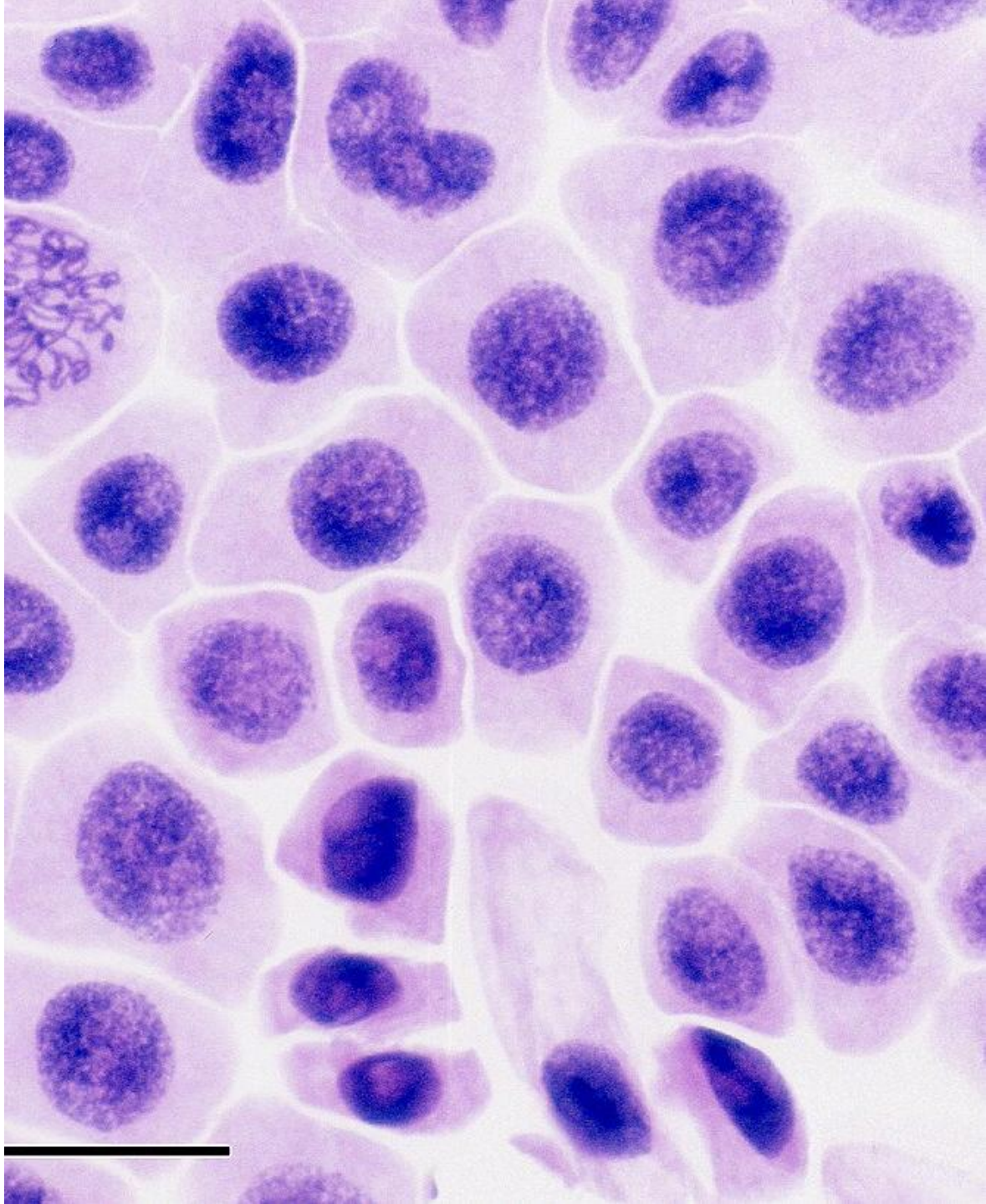
DP Biology – D2.1 (HL extension)

# The cell cycle and cancer

- How are large numbers of cells with identical genes produced?
- How do eukaryotes make genetically varied cells that can become gametes?

HL statements D2.1.12–17. Textbook pp. 662–669.
Image: dividing cells in an onion root tip. CC BY-SA 3.0.



# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | D2.1.12–14 | Explain why cells proliferate, and describe the cell cycle and growth in interphase |
| **2** | D2.1.15 | Explain how rising and falling cyclins control the cell cycle |
| **3** | D2.1.16–17 | Explain how mutations in cell cycle genes cause cancer, and compare benign and malignant tumours |

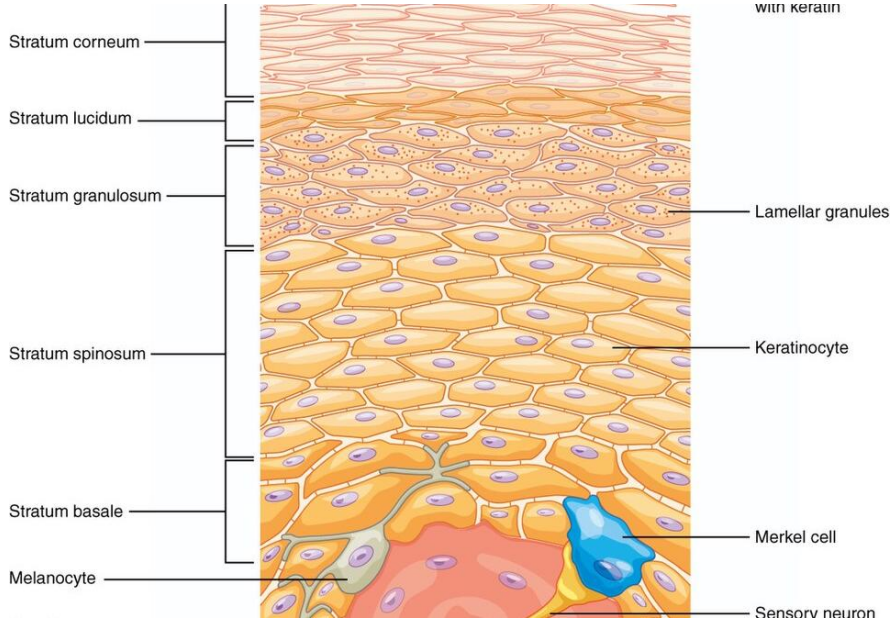
# Why cells proliferate



| Growth | Replacement | Repair |
| --- | --- | --- |
| Plant meristems at shoot and root tips; early embryos. | Skin cells from the basal layer replace those lost at the surface. | Closing a wound. |

## Key idea

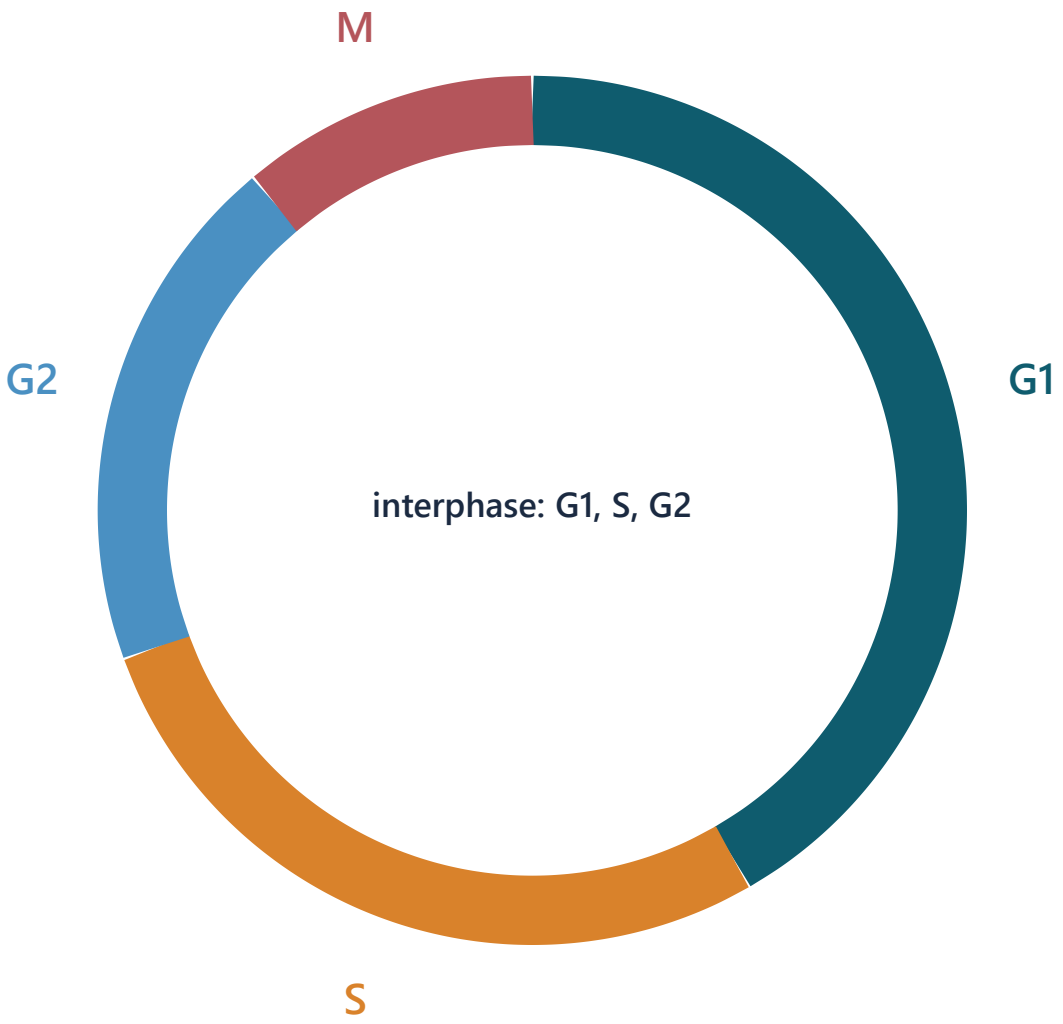
**Cell proliferation** is a fast rise in cell number by repeated cell cycles. Too little and tissues are not repaired; too much and tumours form.




Layers of the epidermis. OpenStax, CC BY 3.0.

# The cell cycle

D2.1.13




| Phase | What happens |
|---|---|
| G1 | Cell grows; proteins and organelles made |
| S | DNA replicated |
| G2 | More growth; preparation for division |
| M | Mitosis, then cytokinesis |

**G0**

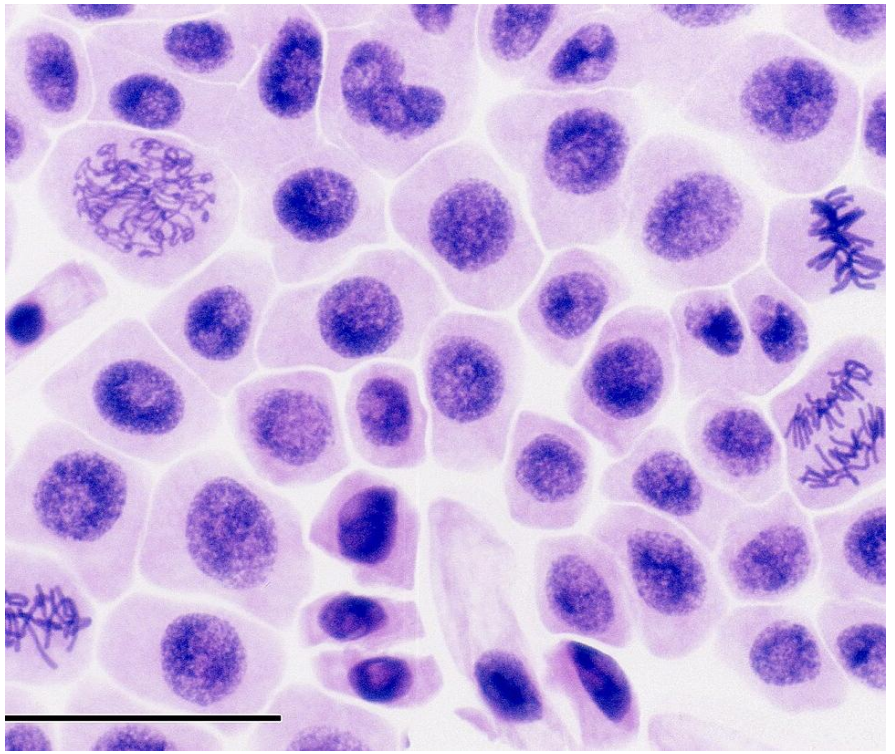
Some cells leave the cycle and stop dividing: for a while (liver) or for good (most nerve cells).

# Interphase is a busy time

D2.1.14

Interphase is **metabolically active**: the cell makes proteins, enzymes and membranes, and replicates DNA in S.

**Mitochondria and chloroplasts** grow and divide, so each daughter gets enough. Ribosomes are made in the nucleolus.



Onion root tip: most cells are in interphase. CC BY-SA 3.0.

### Common mistake

Interphase is not a resting stage; it is the longest and busiest part of the cycle.

# Working with data: time in each phase

D2.1.13–14

Cells counted in an onion root tip photo. Cycle length 24 hours. The values are invented for practice; the pattern is realistic.

| Phase | Cells counted |
|---|---|
| **Interphase** | 180 |
| **Prophase** | 10 |
| **Metaphase** | 4 |
| **Anaphase** | 2 |
| **Telophase** | 4 |

1. Calculate the mitotic index.
2. Estimate the time spent in interphase.
3. Explain the method used for question 2.

# Practice: which phase?

D2.1.12–14

**Doubled DNA**

A cell has twice the normal DNA but no visible chromosomes.

**Never divides**

A mature nerve cell.

**Growing fast**

A cell making many mitochondria and enzymes before replication.

**For each one**

Name the phase and explain.

# Where proliferation happens

D2.1.12

| Place | Why cells proliferate there |
|---|---|
| **Plant root and shoot tips** | Meristems: growth in length |
| **Early embryo** | Builds the body from a single cell |
| **Bone marrow** | Replaces blood cells (red cells live about 120 days) |
| **Skin basal layer** | Replaces cells worn from the surface |

**Think**

Why do cancer treatments often cause hair loss and low blood counts?

# Check-in: class 1

D2.1.12–14

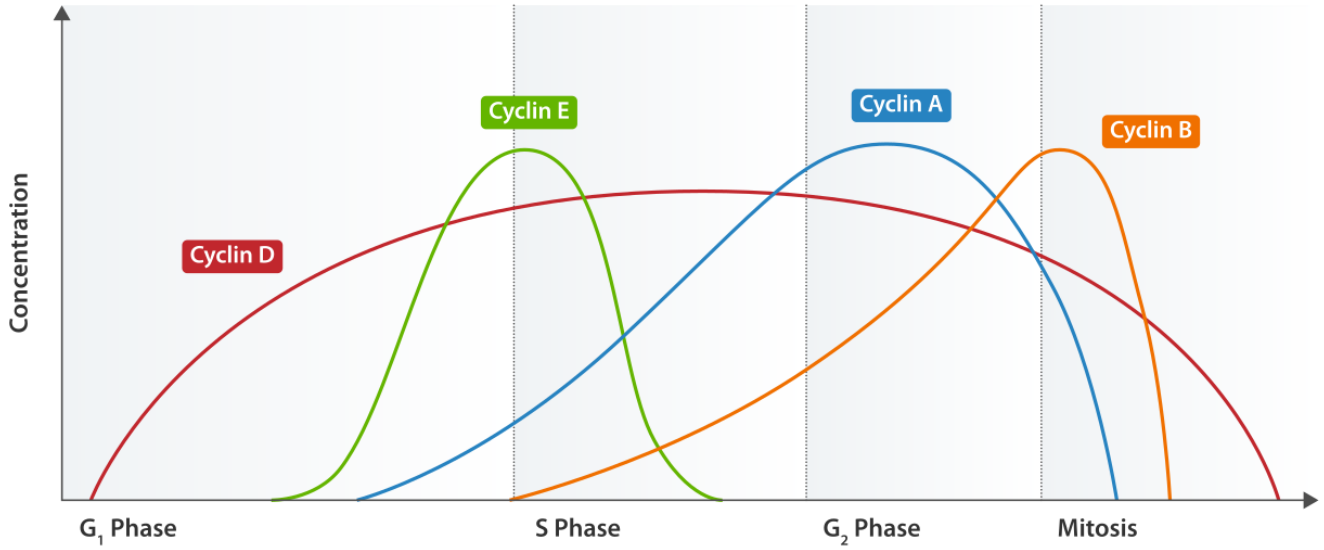
1. Give three reasons for cell proliferation.
2. Name the phases of the cell cycle in order.
3. Outline what happens in G1, S and G2.
4. Define G0.
5. Explain why interphase is not a resting stage.

**How to use this**

Answer without your notes, then check against slides 3–8.

# Cyclins control the cycle

D2.1.15




Concentrations of four cyclins through the cell cycle. Public domain.

Cyclin concentrations **rise and fall** in each cycle. Each cyclin binds and activates a **cyclin-dependent kinase** (CDK).

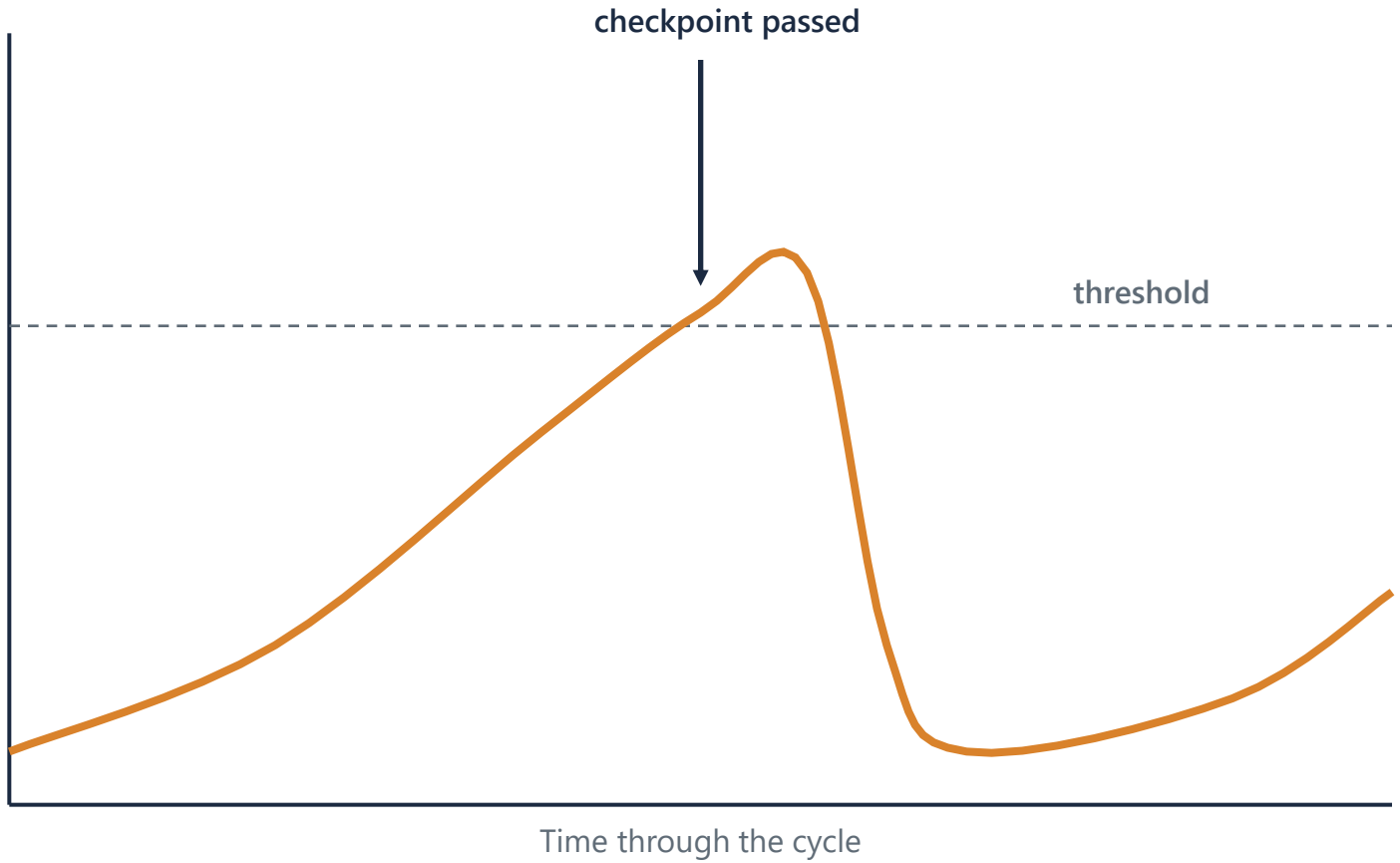
A cyclin must reach a **threshold** for the cell to pass each checkpoint; then it is broken down.

## History

Cyclins were found in sea urchin eggs; the discovery won a Nobel Prize in 2001.

# Reaching the threshold

D2.1.15




The cyclin builds up; once it passes the threshold it activates its CDK, which **phosphorylates** proteins that start the next phase.

Then the cyclin is destroyed, resetting the switch.

# Checkpoints

D2.1.15

| G1 checkpoint | G2 checkpoint | Mitosis checkpoint |
| --- | --- | --- |
| Is the cell big enough? Are conditions right? | Was DNA replicated correctly and fully? | Are all chromosomes attached to the spindle? |

**If not**

The cycle pauses until the problem is fixed, or the cell destroys itself. This stops damaged cells dividing.

# Practice: cyclin problems

D2.1.15

**No breakdown**

Cyclin B cannot be destroyed after mitosis starts.

**Never enough**

A cell cannot make enough G1 cyclin.

**Damage**

DNA damage is found at the G2 checkpoint.

**For each one**

Predict what happens to the cell cycle and explain.

# Check-in: class 2

D2.1.15

1. State what cyclins bind to.
2. Explain how a cyclin moves a cell past a checkpoint.
3. Explain why cyclins must be broken down.
4. Describe one checkpoint.
5. Explain why checkpoints protect the body.

**How to use this**

Answer without your notes, then check against slides 10–13.

# Accelerators and brakes

D2.1.16

**Proto-oncogene**

The accelerator: stimulates division when signalled.

**Oncogene**

A mutated proto-oncogene: accelerator stuck on.

**Tumour suppressor**

The brake: stops the cycle, e.g. if DNA is damaged.

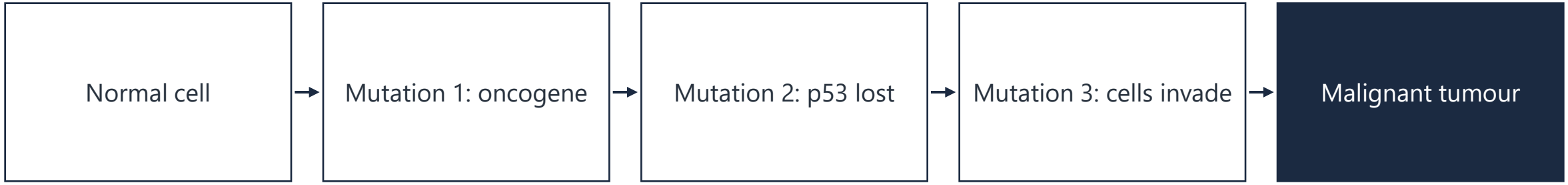
| | Oncogene | Tumour suppressor |
|---|---|---|
| **Mutation** | Permanently on | Stops working |
| **Copies needed** | One | Usually both |
| **Example** | Ras | p53 |

**Result**

Usually several mutations are needed before cells divide out of control.

# Why cancer needs several mutations

D2.1.16




Each mutation gives the cell a small advantage; its descendants collect more. This takes years, so cancer is more common with **age** and with exposure to **mutagens**.

**Link**

Mutagens: tobacco smoke, UV, X-rays (D1.3). Some viruses also carry oncogenes.

# Working with data: smoking and lung cancer

D2.1.16

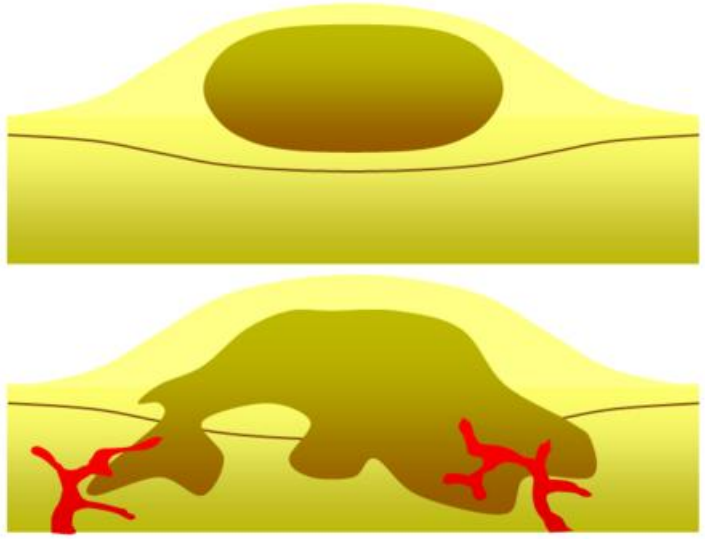
Lung cancer deaths per 100 000 men per year by cigarettes per day (rounded values from a classic study of British doctors).

| Cigarettes per day | Deaths per 100 000 |
|---|---|
| **0** | 10 |
| **1 to 14** | 78 |
| **15 to 24** | 127 |
| **25 or more** | 251 |

1. Describe the relationship.
2. How many times greater is the risk for the heaviest smokers?
3. Explain how smoking can lead to cancer.
4. Why does other evidence point to causation?

# Benign and malignant tumours

D2.1.17



Benign (top) and malignant (bottom). CC BY-SA 3.0.

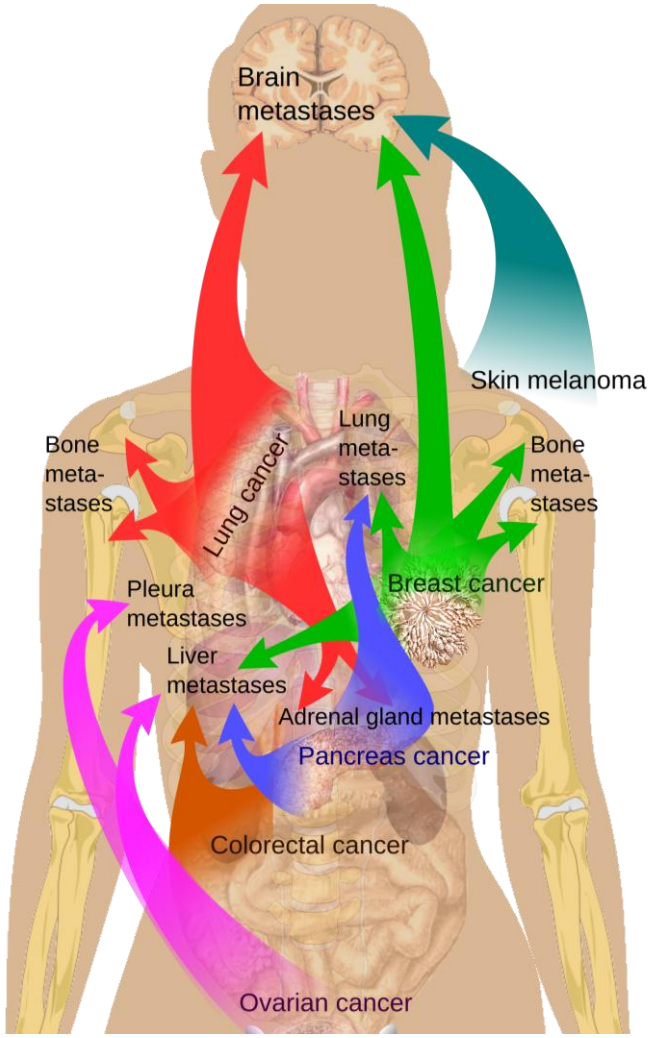
| | Benign | Malignant |
|---|---|---|
| **Growth** | Slow | Fast |
| **Edges** | Contained, often a capsule | Invades nearby tissue |
| **Spread** | Stays in one place | Metastasis |
| **Cancer?** | No | Yes |

**Note**

Benign tumours can still be dangerous if they press on organs such as the brain.

# Metastasis






Common sites of metastasis. M. Haggstrom, CC0.

Malignant cells lose their grip on neighbours, enter blood or lymph and travel. Some settle in new organs and form **secondary tumours**; the first is the **primary** tumour.

Common sites: lungs, liver, bone, brain.

**Think**

Why is cancer much harder to treat once it has spread?

# Practice: explain the case

D2.1.16–17

**Two copies**

Why must both copies of p53 usually be mutated?

**One copy**

Why is one mutated copy of Ras enough?

**Capsule**

A tumour has a clear capsule. Benign or malignant?

**Explain**

Give a reason for each answer.

# Working with data: cancer and age

D2.1.16

New cancer cases per 100 000 people per year. Values invented for practice; the pattern is realistic.

| Age / years | Cases |
|---|---|
| **20–29** | 40 |
| **40–49** | 180 |
| **60–69** | 900 |
| **80+** | 2200 |

1. Describe the trend.
2. Explain it using oncogenes and tumour suppressors.
3. Suggest why childhood cancers are often different.

# Check-in: class 3

D2.1.16–17

1. Distinguish between a proto-oncogene and an oncogene.
2. State the normal role of a tumour suppressor gene.
3. Explain why several mutations are usually needed for cancer.
4. Distinguish between benign and malignant tumours.
5. Define primary and secondary tumours.

**How to use this**

Answer without your notes, then check against slides 15–21.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Interphase is a resting phase.** | It is busy: growth, synthesis, DNA replication. |
| **Cancer is caused by one mutation.** | Several mutations are usually needed. |
| **All tumours are cancer.** | Only malignant tumours are cancers. |

# Summary: the cell cycle and cancer

D2.1.12–17

| | Key idea |
|---|---|
| **D2.1.12** | Proliferation for growth, replacement and repair |
| **D2.1.13–14** | G1, S, G2, M; interphase is active |
| **D2.1.15** | Cyclins activate CDKs at checkpoints |
| **D2.1.16** | Oncogenes and tumour suppressor mutations |
| **D2.1.17** | Benign v malignant; metastasis |

# Check that you can...

1. **Outline** the reasons for cell proliferation
2. **Describe** the cell cycle and events of interphase
3. **Explain** control of the cell cycle by cyclins
4. **Explain** how mutations in cell cycle genes cause cancer
5. **Distinguish** between benign and malignant tumours

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. List the three stages of interphase in order.
2. Explain how cyclins control progress through the cell cycle.
3. Distinguish between a benign and a malignant tumour.

# Image credits

Mitosis (261 13) Pressed; root meristem of onion (cells in prophase, metaphase, anaphase, telophase).jpg – Doc. RNDr. Josef Reischig, CSc. – CC BY-SA 3.0

502 Layers of epidermis.jpg – OpenStax College : J. Gordon Betts, Peter Desaix, Eddie John – CC BY 3.0

Cyclin Expression.svg – Cyclinexpression_waehrend_Zellzyklus.png: Original uploader – Public domain

BENIGN AND MALIGNANT TUMOUR.png – Simon Caulton – CC BY-SA 3.0

Metastasis sites for common cancers.svg – Mikael Häggström – CC0

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